Curve counting on toric Calabi-Yau threefolds: a short summary of the invariants

A note accompanying the Python package `topvertex`

1 Introduction and summary

The package `topvertex` takes a lattice polygon $P \subset \mathbb{R}^2$ with a unimodular triangulation T and returns a power series $Z_T(q, Q) = 1 + \ldots$, together with a table of integers n_β^g (see the main notes [1] for the algorithm). In this note we summarize, informally and without proofs, what these quantities mean. The triangulated polygon encodes a complex 3-dimensional manifold X_T, a *toric Calabi-Yau threefold*, and Z_T is a generating function of numbers that "count" holomorphic curves in X_T. The different ways to make sense of such a count lead to the Gromov-Witten (GW), Gopakumar-Vafa (GV), Donaldson-Thomas (DT) and Pandharipande-Thomas (PT) invariants, listed in Table 1 (see [2] for a review). For a toric Calabi-Yau threefold all four sets of numbers turn out to carry the same information, and Z_T is their common generating function. A change of the triangulation of a fixed polygon changes the invariants in a simple and explicit way, given by the formula (7) below. The geometry of X_T is reviewed in Section 2, the invariants and their relations are described in Sections 3 and 4, and a change of the triangulation is considered in Section 5.

invariant	labelled by	values	what is "counted"	invariant of
GW, $N_{g,\beta}$	$g \geq 0$, β	$\mathbb{Q}$	holomorphic maps $\Sigma_g \to X$	symplectic manifold (X, ω)
GV, n_β^g	$g \geq 0$, β	$\mathbb{Z}$	BPS states of wrapped M2-branes	up to deformations
DT, $I_{n,\beta}$	$n \in \mathbb{Z}$, β	$\mathbb{Z}$	curves and points cut out by equations	complex manifold X
PT, $P_{n,\beta}$	$n \in \mathbb{Z}$, β	$\mathbb{Z}$	curves with points on them	up to deformations

Table 1: The curve counting invariants of a Calabi-Yau threefold X, with $0 \neq \beta \in H_2(X, \mathbb{Z})$.

2 The threefold behind the picture

In this section we review how a triangulated polygon determines a threefold, and which of its features depend on the polygon only. Each lattice point $\omega = (x, y)$ of P is lifted to $\widehat{\omega} = (x, y, 1) \in \mathbb{Z}^3$. The cones over the triangles of T form a fan, and X_T is the corresponding toric variety, with an action of the torus $(\mathbb{C}^*)^3$. Unimodularity of the triangles means that X_T is smooth. The fact that all $\widehat{\omega}$ lie in one plane means that X_T has a nowhere vanishing holomorphic 3-form, which is the *Calabi-Yau* condition, and forces X_T to be non-compact. The triangulation is regular precisely when X_T is quasi-projective, in which case the heights determine a Kähler class. The triangles correspond to the fixed points of the torus action, and the interior segments τ to compact holomorphic curves $C_\tau \cong \mathbb{P}^1$. Polygons related by an affine transformation preserving the lattice $\mathbb{Z}^2$ give isomorphic threefolds.

The following data are the same for all unimodular triangulations of a given polygon P. First, the group of curve classes: $H_2(X_T, \mathbb{Z}) \cong L := \{l \in \mathbb{Z}^N : \sum_i l_i \widehat{\omega}_i = 0\}$, where N is the number of lattice points of P, and the class of C_τ is the relation vector l_τ of [1]. Second, the Euler characteristic $\chi(X_T)$, which is equal to the number of triangles, i.e. to twice the area of P. Finally, the singular affine toric variety X_P of the cone over P, for example the conifold $\{xy = zw\} \subset \mathbb{C}^4$ when P is the unit square. Each X_T is a *crepant resolution* of X_P, that is a resolution of the singularities with a

trivial canonical bundle. What does depend on T is the set of *effective* classes, i.e. the classes of holomorphic curves, which are the non-negative integral combinations of the l_τ.

3 The invariants

In this section we describe the invariants one by one. Let X be a Calabi-Yau threefold.

Gromov-Witten invariants. Informally, $N_{g,\beta}$ is the number of holomorphic maps $f : \Sigma_g \to X$ from Riemann surfaces of genus g, such that the image has class β, counted up to reparametrizations of Σ_g. For a Calabi-Yau threefold such maps are expected to be isolated for all g and β, which is why one obtains numbers without imposing any conditions on the curves. A map with a finite group of automorphisms is counted with the weight $1/|\mathrm{Aut}|$, and therefore the numbers are rational. The GW invariants are invariants of the symplectic manifold (X, ω). They do not change under continuous deformations of the symplectic form ω and of the compatible (almost) complex structure. In particular, for a Kähler manifold they depend neither on the complex structure nor on the Kähler class, as long as these are varied without making X singular.

The topological string. Physically, the GW invariants are computed by the A-model topological string theory on X, a theory of maps from 2-dimensional surfaces to X in which the path integral reduces to a sum over holomorphic maps (see [3] for a review). Its partition function is

$$Z = \exp\Big(\sum_{g\geq 0} g_s^{2g-2} F_g(Q)\Big), \qquad F_g(Q) = \sum_{\beta\neq 0} N_{g,\beta}\, Q^\beta, \tag{1}$$

where g_s is the string coupling and $Q^\beta = e^{-t_\beta}$, with t_β being the (complexified) area of a curve of class β. The theory is sensitive to the Kähler class of X only through the variables Q, and is not sensitive to the complex structure. The series computed by `topvertex` is (1) for $X = X_T$ with $q = e^{ig_s}$. The topological vertex was derived in [4] from a duality with Chern-Simons theory, and was proved to agree with the mathematical definition of the GW invariants in [5, 6]. Since X_T is non-compact, its invariants are in general *defined* by localization, i.e. as sums of contributions of the curves fixed by the torus action.

Gopakumar-Vafa invariants. In M-theory on $X \times \mathbb{R}^5$ an M2-brane wrapping a holomorphic curve of class β is a charged particle in five dimensions. The number n_β^g is a signed count of the supersymmetric (BPS) states of such particles, where g labels their spin content. It was argued in [7] that

$$\log Z = \sum_{\beta\neq 0}\sum_{g\geq 0}\sum_{k\geq 1} n_\beta^g\, \frac{(-1)^{g-1}}{k}\, [k]^{2g-2}\, Q^{k\beta}, \qquad [k] := q^{k/2} - q^{-k/2}. \tag{2}$$

Mathematically, (2) can be taken as the definition of n_β^g in terms of $N_{g,\beta}$, and this is how `topvertex` computes them. The non-trivial statements are then that n_β^g are integers and that $n_\beta^g = 0$ for large g at fixed β. This was proved for toric Calabi-Yau threefolds in [8, 9], and the integrality for general symplectic Calabi-Yau manifolds of real dimension 6 in [10]. Informally, n_β^0 is the number of rational curves of class β with the contributions of multiple covers removed. Being a reorganization of the GW invariants, the GV invariants are invariant under the same deformations.

Donaldson-Thomas and Pandharipande-Thomas invariants. Here a curve is described not by a map but by equations. The integer $I_{n,\beta}$ is a virtual number of 1-dimensional subschemes $S \subset X$ of class β with the holomorphic Euler characteristic $\chi(\mathcal{O}_S) = n$ [11]. Such S is a curve, possibly with multiplicities, together with a finite number of points, which can lie on the curve or away from it. The PT invariants $P_{n,\beta}$ similarly count *stable pairs* [12], which informally are curves with a finite number of points lying on them. We denote the generating functions by $Z_{\mathrm{DT}}(p,Q) = \sum_{n,\beta} I_{n,\beta}\, p^n Q^\beta$ and $Z_{\mathrm{PT}}(p,Q) = \sum_{n,\beta} P_{n,\beta}\, p^n Q^\beta$, where the sums include $\beta = 0$. Both are invariants of the complex manifold X, unchanged under deformations of its complex structure, and no choice of a Kähler class enters the definition. For $X = X_T$ the subschemes fixed by the torus action are described by 3-dimensional partitions (piles of boxes) placed at the triangles [13, 11], and the vertex $C_{\lambda_1\lambda_2\lambda_3}(q)$ is, up to normalization, the generating function of the 3-dimensional partitions with the asymptotic shapes $\lambda_1, \lambda_2, \lambda_3$.

4 Relations

In this section we list the relations between the invariants. Taken together, for $X = X_T$ they state that

$$Z_T(q,Q) = \exp\Big(\sum_{g\geq 0} g_s^{2g-2} F_g(Q)\Big) = Z_{\mathrm{PT}}(p,Q) = \frac{Z_{\mathrm{DT}}(p,Q)}{M(q)^{\chi(X_T)}}, \qquad q = e^{ig_s} = -p, \tag{3}$$

where $M(q) := \prod_{k\geq 1}(1-q^k)^{-k}$. The last equality is the DT/PT correspondence, conjectured in [11, 12] and proved for projective Calabi-Yau threefolds in [14]. The denominator is the part of Z_{DT} with $\beta = 0$, which counts points. The middle equality is the GW/DT correspondence conjectured in [11] and proved for toric threefolds in [6]. The GW and GV invariants can be recovered from each other directly from (2), for example $N_{0,\beta} = \sum_{k|\beta} n^0_{\beta/k}/k^3$. Finally, exponentiating (2) one obtains the product formula

$$Z_T(q,Q) = \prod_{\beta\neq 0}\Big[\prod_{k\geq 1}\big(1-q^kQ^\beta\big)^{k\, n^0_\beta} \prod_{g\geq 1}\prod_{j=0}^{2g-2}\big(1-q^{g-1-j}Q^\beta\big)^{(-1)^{g+j}\binom{2g-2}{j} n^g_\beta}\Big], \tag{4}$$

which relates the GV invariants to the PT invariants. For the resolved conifold, with the only non-zero invariant $n^0_{[C]} = 1$, it reads

$$Z_T(q,Q) = \prod_{k\geq 1}\big(1-q^kQ\big)^k. \tag{5}$$

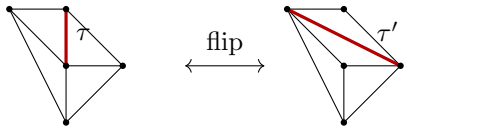
Note that, unlike (2), the GW/DT correspondence is not a relation between individual numbers. For a fixed β, the coefficient of Q^β in Z_{PT} is the series $\sum_n P_{n,\beta}\, p^n$ with finitely many negative powers of p. It is the expansion at $p = 0$ of a rational function of p, invariant under $p \to 1/p$. The GW invariants are obtained by substituting $p = -e^{ig_s}$ into this rational function and expanding it in g_s, that is by expanding it at a different point, $p = -1$. Therefore each $N_{g,\beta}$ depends on infinitely many of the integers $P_{n,\beta}$. For example, the coefficient of Q in (5) is

$$\sum_{n\geq 1}(-1)^{n-1} n\, p^n = \frac{p}{(1+p)^2} = \frac{1}{\big(2\sin\frac{g_s}{2}\big)^2} = \frac{1}{g_s^2} + \frac{1}{12} + \frac{g_s^2}{240} + \ldots, \tag{6}$$

so that $P_{n,[C]} = (-1)^{n-1} n$ for $n \geq 1$, while $N_{0,[C]} = 1$, $N_{1,[C]} = 1/12$, $N_{2,[C]} = 1/240$.

5 Change of the triangulation

In this section we describe how the invariants depend on the triangulation T of a fixed polygon P. Any two unimodular triangulations of P are related by a sequence of *flips* [15]. A flip replaces a diagonal τ of a parallelogram formed by two adjacent triangles with the other diagonal τ' (Figure 1). An interior segment τ can be flipped precisely when its framing in the sense of [1] is zero, that is when C_τ has the normal bundle $\mathcal{O}(-1)\oplus\mathcal{O}(-1)$. Geometrically, a flip is a *flop*. The curve C_τ is contracted to a point, which produces a conifold singularity, and the singularity is then resolved in the other way, by a new curve $C_{\tau'}$. Since X_T and $X_{T'}$ are not related by a continuous deformation through smooth manifolds, the invariance properties of Section 3 do not apply, and the invariants do change. With both groups of curve classes identified with L as in Section 2, we have $l_{\tau'} = -l_\tau$, so that the new curve formally has a negative area in terms of the old variables.



T: local F_1 $\qquad$ T': local $\mathbb{P}^2$ with a curve

Figure 1: The two unimodular triangulations of the polygon with corners $(1,0),(0,1),(-1,1),(0,-1)$. On the left, $l_\tau = B$ is the class of the base of the Hirzebruch surface F_1. On the right, the class of a line in $\mathbb{P}^2$ is $H = B + F$, where F is the class of the fiber of F_1, and the class of the extra curve is $-B$.

The change is, however, rather mild. Namely, for regular triangulations T and T' related by a flip of τ one has

$$n^g_\beta(X_{T'}) = n^g_\beta(X_T) \qquad \text{for all } g \text{ and all } \beta\in L \text{ not proportional to } l_\tau, \tag{7}$$

while the only non-zero invariant on the line spanned by l_τ is $n^0_{l_\tau}(X_T) = 1$, which is replaced by $n^0_{-l_\tau}(X_{T'}) = 1$. Equivalently, in terms of the product formula (4),

$$\frac{Z_{T'}(q,Q)}{\prod_{k\ge1}(1-q^kQ^{-l_\tau})^k} = \frac{Z_T(q,Q)}{\prod_{k\ge1}(1-q^kQ^{l_\tau})^k}\,. \tag{8}$$

Here both sides are power series in the monomials Q^β with β effective both in X_T and in $X_{T'}$. In view of (3), the same statement holds for the GW, PT and DT invariants. For the topological vertex the formula (8) was proved in [16]. It is a special case of general results, which do not require X to be toric: for the GW invariants in all genera [17], and for the DT and PT invariants [18, 19]. Mathematically, all X_T have equivalent derived categories of coherent sheaves, and the invariants of X_T and $X_{T'}$ count objects of this category that are stable in two different senses [18, 19].

Example: local $\mathbb{P}^1$. The simplest example is the unit square with its two triangulations T and T', given by the two diagonals. Both X_T and $X_{T'}$ are the resolved conifold, i.e. the total space of $\mathcal{O}(-1)\oplus\mathcal{O}(-1)\to\mathbb{P}^1$, but the classes of the two curves are opposite generators of $L\cong\mathbb{Z}$. With $Q := Q^{l_\tau}$, the formula (5) gives

$$Z_T(q,Q) = \prod_{k\ge1}\bigl(1-q^kQ\bigr)^k, \qquad Z_{T'}(q,Q) = \prod_{k\ge1}\bigl(1-q^kQ^{-1}\bigr)^k, \tag{9}$$

and both sides of (8) are equal to 1. The two series in (9) are expansions at $Q=0$ and at $Q=\infty$, and it is instructive to compare them as functions of Q. From (2) and (6), their free energies in

the genus expansion (1) are

$$F_g(Q) = c_g \operatorname{Li}_{3-2g}(Q), \qquad F'_g(Q) = c_g \operatorname{Li}_{3-2g}(1/Q), \qquad \operatorname{Li}_s(Q) := \sum_{k\geq 1} \frac{Q^k}{k^s}, \tag{10}$$

where $c_0 = 1$, $c_1 = 1/12$, $c_2 = 1/240, \ldots$ are the coefficients in (6). For $g \geq 2$ the function $\operatorname{Li}_{3-2g}(Q)$ is a rational function of Q invariant under $Q \to 1/Q$, for example $\operatorname{Li}_{-1}(Q) = Q/(1-Q)^2$. Therefore F_g and F'_g are expansions of the same function. For $g = 0, 1$ this is true only up to a polynomial in $\log(-Q)$:

$$\operatorname{Li}_3(Q) - \operatorname{Li}_3(1/Q) = -\frac{1}{6}\log^3(-Q) - \frac{\pi^2}{6}\log(-Q), \qquad \operatorname{Li}_1(Q) - \operatorname{Li}_1(1/Q) = -\log(-Q). \tag{11}$$

A cubic polynomial in the Kähler parameter $t = -\log Q$ in genus 0, and a linear one in genus 1, have the form of the contributions of $\beta = 0$ to the free energy, which are not included in Z_T. For a compact Calabi-Yau threefold these contributions are given by the triple intersection numbers and by the second Chern class, which change under a flop in a way that compensates (11) [20, 21]. Physically, the Kähler cones of the different X_T are glued along their walls into a single space of Kähler parameters, and the complete free energy is analytic across the walls. In this sense, in the general formula (8) the factors of the flopped curve also match, and Z_T and $Z_{T'}$ are expansions of the same function in two different regions.

Example: local F_1. For the polygon in Figure 1, the relation (7) implies that the invariants of local F_1 in the classes $dB + dF$ coincide with the invariants $3, -6, 27, \ldots$ of local $\mathbb{P}^2$ in degree d. We verified (7) with `topvertex` for this polygon, for all classes up to a certain weight, including $g > 0$.

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